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Reproduction and embryonic development of monkeyface prickleback *Cebidichthys violaceus* in captivity

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Abstract

Understanding the sequence of embryonic and larval development and the factors necessary to induce reproduction in captivity are critical for developing new species for commercial or conservation aquaculture. In this study, we describe the adult reproductive behaviours and development of eggs, embryos and early larvae of captive monkeyface pricklebacks, *Cebidicthys violaceus*, compared to previously documented wild observations. Eggs were laid in cohesive clutches and guarded by the male parent until hatching began 23 days post fertilization at 13°C. Fertilized eggs were spherical, approximately 1.5 mm in diameter, covered in an opaque chorion, and contained six adhesive pads around the outside. We characterized the rate of depletion of yolk and the oil globule and growth of the embryo from fertilization until hatching. Notable embryonic stages were documented, including the timing of the first heartbeat, and the development of otoliths, intestinal tract, eye pigmentation, mouth, fins and the circulatory system. Larval length at hatching was about 7.4 mm, and larvae were immediately mobile and feeding on live food. Larvae were cultured and observed up to 18 days post hatch.

KEYWORDS
Cebidichthys violaceus, embryo, larvae, monkeyface prickleback, reproduction

1 | INTRODUCTION

The family Stichaeidae (pricklebacks) contains about 38 genera and 81 species found throughout temperate marine waters of the North Pacific, Arctic and North Atlantic Oceans (Nelson et al., 2016). Pricklebacks tend to be small (<30 cm), elongate, benthic fishes inhabiting the intertidal zone and shallow bays to depths of 250 m or more (Mecklenburg & Sheiko, 2004). Little is known about the reproductive biology of this family overall, although observations of several species indicate they are oviparous and have adhesive eggs that are guarded by at least one parent (Coleman, 1992; Coleman et al., 1999; Marliave & DeMartini, 1977; Wourms & Evans, 1974a).

The monkeyface prickleback, *Cebidichthys violaceus* (Girard 1854), is the largest observed species of prickleback, growing up to 76 cm in length and 2.3 kg (Marshall & Echeverria, 1992). This species inhabits rocky intertidal and shallow subtidal areas from Oregon to northern Baja California, where it is commonly fished by recreational anglers. Monkeyface pricklebacks are herbivorous, feeding on a diversity of red and green macroalgae (Barton, 1982; Burge & Schultz, 1973), with a diet that shifts seasonally (German et al., 2004; Horn et al., 1982), often with algal availability in the intertidal. Like many other pricklebacks, the reproductive biology and early life history of *C. violaceus* remain largely unknown. Limited observations of *C. violaceus* eggs have been documented, only denoting that they are demersal with a

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diameter of 2.7 mm, and indications that the eggs are adhesive to rocky substrates (Fitch & Lavenberg, 1971; Marshall & Echeverria, 1992; Miller & Marshall, 1987). Other reproductive traits have been documented, including the phenology of their spawning period (January to August), size on recruitment to the intertidal zone (range 1.7–2.2 cm) and size at maturity (50% of females mature at 39 cm; Marshall & Echeverria, 1992; Miller & Marshall, 1987).

Recently, *C. violaceus* has been identified as a candidate for finfish aquaculture, due in part to its herbivorous feeding habits (Horn et al., 1982). Fishing of forage fish for the extraction of fishmeal and fish oil used in formulated feeds for aquaculture and other livestock is often considered unsustainable due to growing aquaculture agriculture industries and limited stocks of forage fish (Ghamkhar & Hicks, 2020; Takon & Metian, 2008). However, efforts to substitute fish-based ingredients with plant and algae sources in many of the carnivorous fish species used in aquaculture have resulted in nutritional challenges including reduced growth, feed efficiency and fish health, likely because carnivorous species have not evolved the necessary digestive physiology to consume plant and algae material (Hua et al., 2019; Yamamoto et al., 2023). For this reason, the culture of marine herbivorous fishes is likely more suited to fish-free alternative ingredients, as shown in recent efforts to develop culture technology for Rabbitfish (Siganidae) and Sea Chubs (Kyphosidae), which could aid in reducing unsustainable pressures on forage fish populations (Li et al., 2018; Merlo et al., 2025; Thépot et al., 2022). However, successful introduction of a novel species to aquaculture requires knowledge of its reproductive biology to develop appropriate spawning techniques. Additionally, the identification of critical embryonic and larval stages, such as eye and gut formation, time of hatching and first feeding, is essential for successfully rearing any fish species. Prior to this study, captive spawning and rearing have not been documented for *C. violaceus*. Reproductive and early life-history characteristics are mostly unknown, leaving a gap in knowledge that is critical to building aquaculture protocols and understanding the species. The purpose of this investigation was to describe reproductive behaviours and fecundity of adult *C. violaceus*, and to document embryonic and larval developmental features to aid in future attempts to spawn and rear this species in captivity.

2 | METHODS

2.1 | Study species collections and husbandry

Adult *C. violaceus* were collected from multiple locations along the Central California coast, ranging from Half Moon Bay to Monterey Bay, between September 2020 and February 2021. To collect fish, poke pole fishing methods were used, which involved prodding a pole with a baited hook into crevices in intertidal zones at low tide. Seventy-two adults were collected and transported to Moss Landing Marine Laboratories in aerated coolers. Fish were individually measured for total length (L_T) and weight (W), and tagged with Biomark PIT tags for identification throughout the study. To identify the sex of each fish for reproduction experiments, a portable ultrasound machine (GE Healthcare Venue 40) was used, viewing the abdominal area to locate the presence/absence of an ovary. Fish with an observed ovary were identified as female and fish lacking any visible reproductive organs that were above the size at maturation were identified as male. Fish were sedated in a bath of 50 mg/L tricaine methane sulfonate (MS-222) prior to the ultrasound. At the conclusion of experiments, the pricklebacks were humanely euthanized and dissected to visually confirm the sex of each individual through macroscopic examination of the gonads.

Monkeyface pricklebacks were held outdoors in 800-L circular tanks (2.0 m diameter) under shade with exposure to natural photoperiod. The tanks received flow-through seawater from the head of the Monterey Submarine Canyon, subject to natural oceanic variability (9–21°C, dissolved oxygen 40–600 μmol L^{-1}, salinity ~33 PSU). Tanks contained four dens, each made by aligning two cinder blocks on their sides with a concrete cap block cemented to one end. These structures replicated the caves and crevices the fish reside in naturally. Adult *C. violaceus* were fed a mixture of live macroalgae, including the green alga sea lettuce (*Ulva* spp.) and several species of fleshy red algae (dulse [*Develeraea mollis*], nori [*Pyropia* spp.] and *Mazzaella* spp.) known to be consumed by this species in the wild (Edwards & Horn, 1982; Horn et al., 1982). The fish were fed ad libitum every other day, with uneaten algae removed weekly.

2.2 | Adult conditioning and reproduction

We attempted captive spawning starting in February 2021, using nine tanks stocked with four mature males and four mature females each, determined via ultrasound. Egg surveys were conducted visually two to three times per week using a Teslong NTS300 Inspection Camera to observe inside the dens for egg masses. No complete egg masses were observed during this first spawning trial, but there were many instances where unfertilized eggs were found scattered throughout the tank. These results suggested spawning may have occurred by the female fish, but aggressive or competing males or females may have destroyed the egg masses. Eggs were collected and examined for signs of fertilization (i.e. adhesive pads on an opaque chorion) and no eggs were fertilized. In June 2021, presumably after the spawning season concluded, all fish were moved to a communal 3500-L holding tank.

In February 2022, adult *C. violaceus* were returned to the 800-L tanks for a second attempt at spawning in captivity. Based on the lack of successful spawns in 2021 and observations of aggression between males, spawning groups were designed to test the effect of density and sex ratio on spawning success. Four treatment groups (one male:three female, one male:two female, one male:one female, two male:one female) were used with four replicates each, resulting in 16 total spawning groups. Several successful spawns were documented. From June 2022 to April 2024, a single pair of *C. violaceus* was left in a tank with minimal interference. The chosen pair had produced a viable egg clutch in the spring of 2022 and therefore represented a confirmed breeding pair.

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2.3 | Egg sampling and incubation

When found, fertilized egg masses were transferred into 7.5-L egg incubation tanks. Egg incubators were placed on a flow-through system with aeration inside a water bath maintained at 13°C. Five eggs per day were removed from the egg mass and examined under a dissecting microscope to describe the stages of embryological development and growth (e.g. total embryo length, yolk sac volume, oil globule volume, developmental milestones).

To create a developmental time series, embryos were measured, photographed and the developmental stage was determined daily. To sample and measure embryo development, eggs were dechorionated to remove the opaque exterior of the egg and gently extract the living embryo (Iwamatsu, 1983). Using two #5 forceps, the egg was held stationary while the chorion was gently peeled off in pieces until the live embryo could be extracted without being damaged. Embryos were then suspended in Neutral pH (Lineco$^{(R)}$), a pH-neutral crystalline gel, to reduce movement during observations and photographs.

Egg clutches produced by the final pair between June 2022 and April 2024 were left with the parents to observe parental behaviours until eyespots appeared in the eggs.

2.4 | Analyses

Fecundity was determined by dividing the total volume of the egg mass by the average volume of one egg. Egg volume (V_E) was calculated using the volumetric equation of a sphere:

$$V_E = \frac{\left(\frac{4}{3}\right) \times \pi \times \left(\frac{d}{2}\right)^3}{1000}$$

where d is the diameter of the egg. Fecundity was calculated by dividing the egg mass volume by the average volume of a single egg, providing an approximate number of eggs produced in the clutch. To estimate and track energy reserves throughout embryonic and larval development, yolk and oil globule volumes were calculated for every sample taken following methods used in previous studies (Blaxter & Hempel, 1966; Cetta & Capuzzo, 1982). Yolk volume (V_Y) was measured using the equation for a prolate spheroid

$$V_Y = \frac{\pi}{6} L H^2$$

where L and H are the diameters of the longer and shorter axes, respectively. Oil globule volume was measured using the same equation as egg volume.

2.5 | Larval rearing

After hatching, larvae were transferred to 7.5-L incubation tanks. A 1000-μm screen on the outflow prevented larvae from escaping.



FIGURE 1 A male *Cebidichthys violaceus* guarding an egg mass laid in 2024. The tail can be seen wrapped around the mass on the right side of the image.

Larvae were fed live L-type rotifers, *Brachionus plicatilis*, hourly (~10 count mL^{-1}) using peristaltic pumps and *Artemia* sp. nauplii once per day (~3 count mL^{-1}). Live foods were batch cultured using methods outlined by Dhont and Van Stappen (2003) and Kailasam et al. (2015). Larvae were sampled daily to track growth and developmental changes until one larva remained, which was left undisturbed until it expired naturally. All larvae were sampled live and measurements were taken in seawater to avoid shrinkage.

2.6 | Ethics statement

All procedures involving these fish were reviewed and approved by the San José State University Institutional Animal Care and Use Committee (protocol no. 1068) and were performed in accordance with the PHS Policy on Humane Animal Care and Use of Laboratory Animals administered by the Office of Laboratory Animal Welfare/NIH (Institutional Assurance #D16-00389). Scientific Collection Permits were approved by the California Department of Fish and Wildlife (SCP - S-191050002-19126-001).

3 | RESULTS

3.1 | Adult reproduction

During 2021, 15 egg-release events were observed, eight in April and seven in May. These egg-release events resulted in eggs scattered throughout the enclosure rather than occurring in a tight, adhesive cluster as was expected based on literature reports of wild observations. All eggs were determined to be unfertilized, indicated by the lack of an adhesive chorion and no embryonic development when incubated for 3 days.

In 2022, seven egg-release events occurred, with five of them mirroring those in 2021. Two successful spawning events occurred on

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TABLE 1 Embryological and larval developmental milestones of *Cebidichthys violaceus*.

	Figure 2 panel	Stage	Characteristics
DPF			
0	a	Egg	Fertilized eggs have adhesive pads and a fully developed, opaque chorion
1	b	Gastrula	50% epiboly; blastoderm remained uniform across animal pole
3	c	Gastrula	50% epiboly; shield was visible from animal pole
4	d	Gastrula	100% epiboly; tail bud prominent, first sign of head
5	e	Segmentation	Eight-somite stage; optic vesicle development
6	f	Segmentation	Hindbrain development, Kupfer's vesicle development
7	g	Segmentation	Beginning of tail development
8	h	Segmentation	Tail extended, divisions in hindbrain appeared
9	i	Pharyngula	First heartbeat, first sign of otoliths (Figure 2i′), gut tract began to develop, eye pigmentation appeared
10	j	Pharyngula	Anal and caudal fins begin developing
14	k	Pharyngula	First observation of the "blood island"
16	l	Pharyngula	Start of pectoral fin development (Figure 2l′)
20	m	Pharyngula	Swimming ability, melanophores appear on intestines (Figure 2m′)
22	n	Pharyngula	Mouth and fins fully developed (Figure 2n′)
23	0	Hatching	Chorion degraded (Figure 2o′), first hatching occurred
DPH			
1	p	Larval period	First feeding of rotifers
2	q	Larval period	Yolk sac depleted
3	r	Larval period	First feeding of *Artemia*

Note: Due to the extended hatching period, days post fertilization (DPF) is switched to days post hatching (DPH) for larval development to standardize the timeline. Images showing full body development and highlighted characteristics in certain cases are presented in Figure 2.

8 May, 2022 and 19 May 2022 where large egg clusters were observed inside the dens, pressed as far away from the opening as possible. In both cases, egg masses were observed in the morning with the males holding the egg masses in place by wrapping the posterior half of their body around the clutch and displaying egg-fanning behaviour (Figure 1). The guarding male became aggressive when objects such as cameras or siphon tubes approached the entrance of the den. The males were observed to fan the eggs by wrapping their caudal fin around the egg mass and pushing water over it via movements of the pectoral fins. Fanning of the eggs occurred continuously while the male guarded each nest. In addition, males were uninterested in offered food while guarding egg masses. Eggs of both clutches were light yellow in colour, small and strongly cohesive within the mass but not adhesive to the substrate.

The spawning event on 8 May 2022 occurred in a treatment containing one male ($L_T = 58$ cm) and one female ($L_T = 54$ cm). The average egg diameter was 1.52 (±0.10) mm, with an estimated clutch size of 134,434 eggs. The egg mass was divided into smaller clumps and relocated into incubator buckets on 9 May 2022.

The spawning event on 19 May 2022 occurred in a treatment containing one male ($L_T = 65$ cm) and three females ($L_T = 43$, 46.1, and 52.5 cm). It is unknown which female produced the eggs as the male was providing care and guarding against female intrusion into the den. The average egg diameter was 1.61 (±0.08) mm with approximately 97,753 eggs in the whole clutch. The egg mass was removed the same day it was discovered and transferred into incubator buckets.

No spawning events occurred during 2023. On 21 March 2024 an egg clutch was observed from the only remaining pair of *C. violaceus*, the same pair that spawned on 8 May 2022. This egg clutch was left with the guarding male for as long as possible to further observe parental care, with the intent of removing them shortly before hatching. On 28 March 2024, the eggs were discovered scattered throughout the enclosure, with most of the clutch missing. The few remaining eggs were collected and placed into an incubator bucket. The average diameter of the remaining eggs was 1.63 (±0.07) mm. Because of the scattering of the eggs, clutch size was not calculated, but the eggs were confirmed to be fertilized and developing. No successful hatches occurred in 2024, although many deceased larvae were observed partially hatched, indicating issues with the chorion breaking down.

3.2 | Embryology

Notable developmental characteristics are listed in Table 1 and shown in Figure 2. Fertilized eggs were spherical in shape with six adhesive pads in a cruciate pattern (Figure 2a). At 1 day post fertilization (DPF), the opaque chorion was fully developed. In each egg, the yolk was situated centrally with one oil globule present. This study was unable to document the first cleavage and blastula period, but the gastrulation

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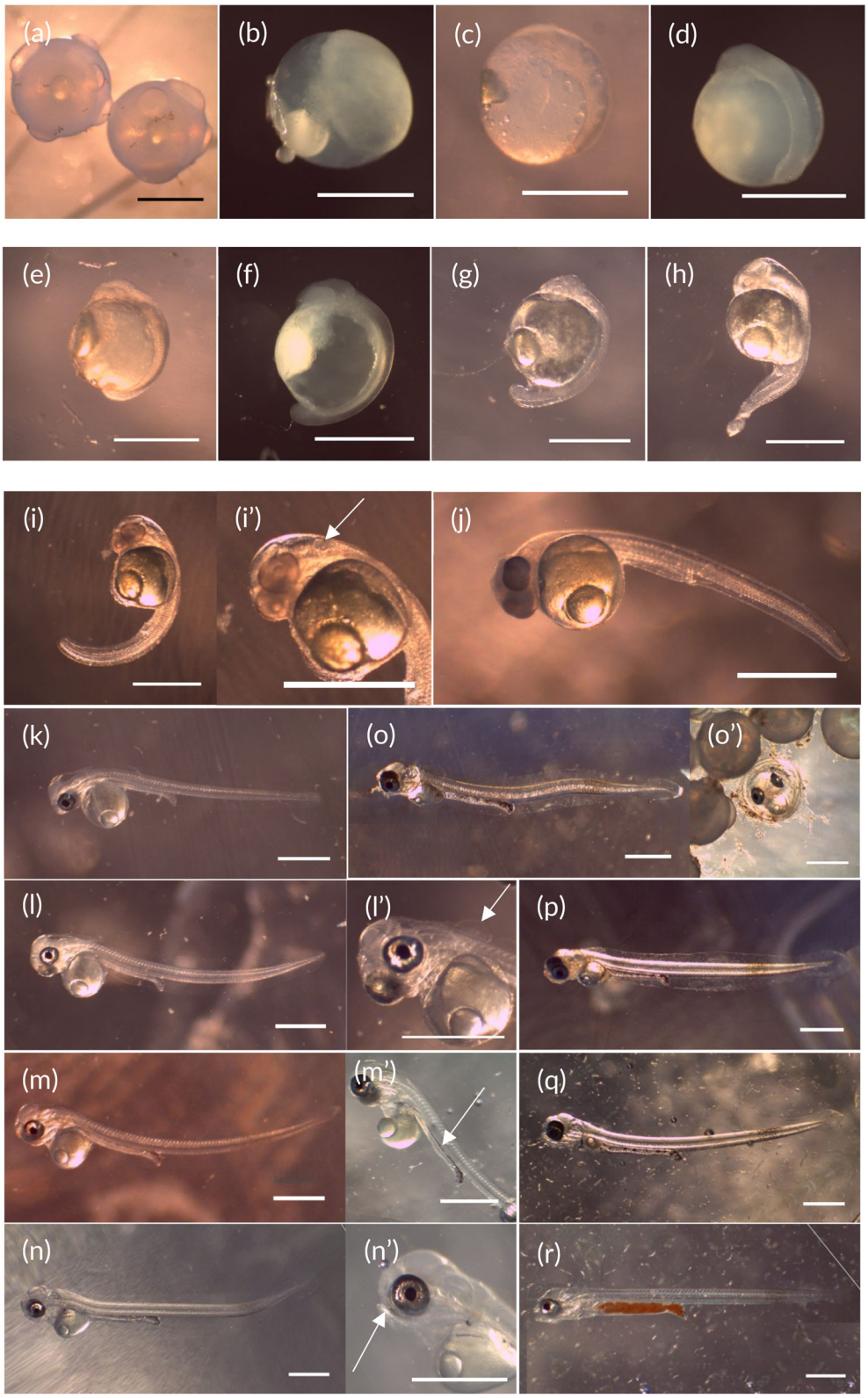


FIGURE 2 Development timeline of *Cebidichthys violaceus* embryos (a–n) and larvae (o–r). Scale bars in the bottom right corner of each image represent 1 mm. Images show key stages of development, including the gastrula stage (b–d), segmentation stage (e–h) and pharyngula stage (i–n). Certain notable features are shown at a higher magnification with arrows including first development of otoliths (i′), pectoral fins (l′), pigment on intestines (m′) and the mouth (n′). o′ displays the breakdown of the chorion immediately prior to hatching compared to eggs that retain the chorion and are opaque. Reference Table 1 for specific days post fertilization, days post hatching and other notable developments for each image.

period (1–4 DPF) was marked by three different stages: the 50% epiboly stage at 1 DPF (Figure 2b), the shield stage at 3 DPF (Figure 2c), and the bud stage at 4 DPF, which occurs when there is 100% epiboly (Figure 2d). During the bud stage swelling occurred on each end of the embryo, which further developed into the head and tail.

At 5 DPF, the segmentation period (5–9 DPF) began where the eight-somite stage became visible and the first development of the optic vesicle was observed (Figure 2e). The first observation of hindbrain development and the beginning development of the Kupfer's vesicle occurred at 6 DPF (Figure 2f). Kupfer's vesicle and the

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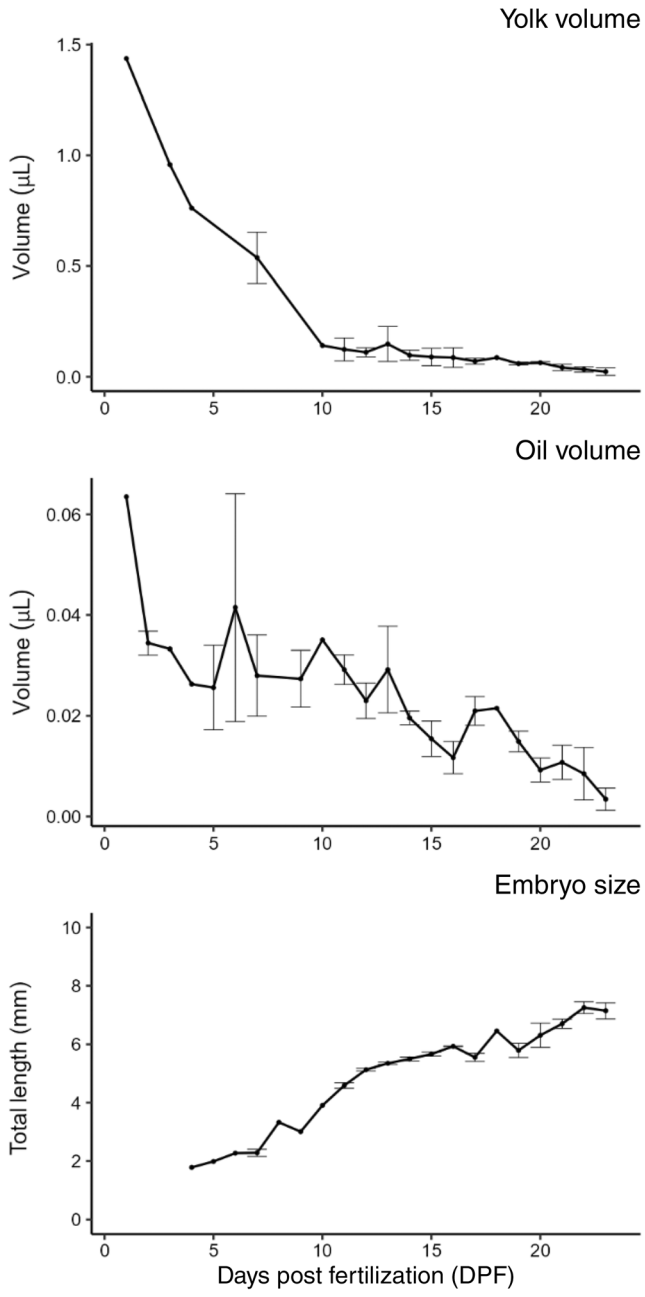


FIGURE 3 Endogenous resource utilization and growth of *Cebidicthys violaceus* embryos. Average volume of yolk (top) and oil globule (middle), and total length (bottom) from fertilization to hatching at 23 days post fertilization. Larvae averaged 7.43 (±0.72) mm at hatching with approximately 1.3% of yolk and 5.1% of oil globule volumes remaining.

optic vesicle continued to develop and by 7 DPF the tail began to develop (Figure 2g). By 8 DPF the tail became well extended from the yolk and there were observable divisions of the hindbrain, creating a hollowed-out area of the otic vesicle. Somites and the notochord continued developing, exhibiting a characteristic stack of pennies appearance along the trunk of the embryo (Figure 2h).

Transitioning from the segmentation to the pharyngula period at 9–10 DPF is a critical period in development. Muscle developed along the notochord with spontaneous myotomal contractions that produced the first movement of the embryo. At this stage, the tail was well extended and otoliths were first observed. Development of the anus, intestine and gut tract was recorded (Figure 2i). Pigment first appeared on the outer eyes at 9 DPF but continued to develop with the eyes until hatching. The circulatory system began to form during this period where the cone-shaped heart was visible and observed to be beating, located directly behind the eyes. The pronephric duct began to form but was difficult to visualize at this stage. Anal and caudal membranes began to form the beginning of larval fins, which were first visible at 10 DPF (Figure 2j).

Throughout the pharyngula period (10–22 DPF) the embryo had a clear body plan and began developing general features such as fins, eyes and a mouth. The circulatory system continued to develop and at 14 DPF the "blood island" (Kimmel et al., 1995) which consisted of the majority of the first dividing and differentiating red blood cells was clearly visible (Figure 2k). These blood cells eventually moved anteriorly to occupy spaces around the yolk ball (18 DPF) and became key parts of the circulatory system. At 16 DPF the pectoral fins began developing (Figure 2l). As fins developed, spontaneous twitching became less common, but embryos became more reactive to external disturbances. At 20 DPF melanophores appeared on the intestines and pigmentation developed in longitudinal stripes on the tail and caudal fin (Figure 2m). By 22 DPF the pectoral fins and mouth were fully developed and locomotion of the tail began, leading to swimming ability before hatching (Figure 2n).

Hatching of exterior eggs first began at 23 DPF and lasted until 31 DPF. Just prior to hatching, the chorion was observed to degrade, allowing the embryo to easily break through the remaining thin membrane (Figure 2o). Eggs near the periphery of the egg mass developed faster and hatched before those in the centre. Because of the high mortality resulting from an unidentified fungal infection on the egg mass, only 108 eggs hatched.

Endogenous energy reserves and embryo size were measured throughout embryological development and growth (Figure 3). Embryos grew steadily during development although growth appeared to slow slightly after the yolk sac was mostly depleted around 12 DPF. The yolk sac was primarily utilized during the first 10 days of embryological development and was fully depleted at 2 days post hatching (DPH). Following the depletion of the yolk reserves, the oil globule volume remained around 50% at 10 DPF and became the main energy source throughout the rest of embryological development until hatching, when both yolk and oil globule reserves were less than 5%.

Newly hatched larvae had an elongated body form, very small yolk and oil globule, and rows of melanophores along the gut and ventral tail region (Figure 4). The mouth was formed and functional, and the eyes were heavily pigmented, but the caudal fin was not differentiated from the dorsal and anal fins. Length at hatching ranged from 6.82 to 8.26 mm with an average of 7.43 (±0.72) mm. Snout to anus length was approximately 44.2% (±1.4) of the total body length. Head length, measured from snout to the operculum edge, was approximately 14.1% (±0.7) of the total body length.

First feeding occurred at 1 DPH, with ingestion of *Brachionus plicatilis* (L-type average 239 µm) and confirmed by the presence of

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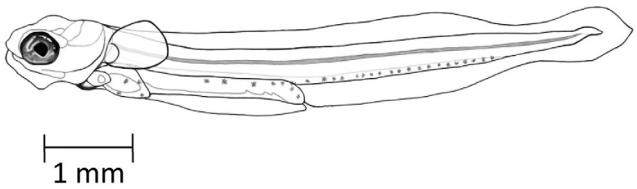


FIGURE 4 Newly hatched larva of *Cebidichthys violaceus*.

green rotifers within the stomachs of larvae that were sampled. Larvae were observed eating a larger food item of *Artemia* (instar I nauplii 300–500 μm) at 3 DPH (Figure 2r). Active hunting and strikes on prey were observed and recorded on video directly after feeding. Because of the high mortality and the destructive nature of sampling, larvae were not sampled beyond 3 DPH. The remaining larvae were cultured until the final larva expired at 18 DPH, which were unable to be sampled due to rapid decomposition of the body.

4 | DISCUSSION

This study was the first to successfully breed *C. violaceus* in a captive setting and, despite some progress, it encountered several challenges in achieving consistent spawning under captive conditions. Out of the 72 broodstock fish utilized in this experiment, only two pairs managed to successfully spawn and fertilize egg masses in captivity. While the success rate was low, the spawning events provide valuable insights into reproductive dynamics. Notably, the common observation of scattered, unfertilized eggs observed in 2021 suggests that interactions between fish, most likely males, led to decreased spawning success and not resource limitation precluding females from producing eggs. This negative interaction between males, while rare, has been reported in other well-studied species such as Nile tilapia and European bitterling (Khater, 2002; Reichard et al., 2004). Male–male competition between *C. violaceus* was observed throughout this study where tanks with multiple males consistently showed indications of fighting (distinct bite marks or injuries to male fish) and contained the majority of egg-scattering events. Additionally, the only successful spawning events occurred in tanks with only a single male present.

Fish behaviour beyond antagonistic male–male interactions such as courtship and mate selection may also have influenced the spawning success of *C. violaceus* in this study. Observations of courtship behaviours proved challenging due to unique behaviours by the species and limited visibility afforded by the tank design used. Observations of multiple (three or four) individuals residing in the same den and frequent switching of dens made it difficult to determine when pairing occurred. Additionally, because *C. violaceus* is largely nocturnal (Ralston & Horn, 1986), courtship behaviour and spawning events were not observed. A noteworthy observation of the egg masses was that the shape of the masses indicated they were pressed into the back of the dens as the masses were being produced. This suggests spawning, and possibly courtship, took place inside the dens and eggs were produced as far from the opening as possible. Interestingly, it was discovered in this study that fertilized eggs are extremely cohesive to each other, but not to the substrate. This result is contrary to previous reports that eggs were adhesive to rocks (Fitch & Lavenberg, 1971), although egg adhesion is largely unknown in Stichaeids.

This study revealed insights, such as parental care provided by the male, which was previously unknown for the species. In all instances of successful spawning, the males were observed diligently guarding and engaging in fanning and care of the egg mass. The identification of uniparental care by the male in *C. violaceus* is consistent with previous studies which report that Stichaeidae are one of the few families known to contain species with either male-only (e.g. *Xiphister mucosus* and *X. atropurpureus*; Marliave & DeMartini, 1977) or female-only (e.g. *Anoplarchus purpurescens*; Coleman, 1992) parental care, but no species with biparental care.

While the spawning experiments conducted in this study did not indicate clear associations with sex ratio, two successful spawning events generating clutches of fertilized eggs provided the first and valuable opportunity to track embryological development of the species. The development of *C. violaceus* embryos was observed to be fairly slow (23–30 days), which is typical of many cold-water species, including cabezon *Scorpaenichthys marmoratus* (25–45 days; Lauth, 1989), lingcod *Ophiodon elongatus* (21–38 days; Gregg, 2003) and the closely related black prickleback *Xiphister atropurpureus* (21 days; Wourms & Evans, 1974b).

The low hatching rate (<1%) in this study is likely due to bacterial or fungal infections of the eggs. While a definitive identification of the growth on the embryos was not determined, the outward white filamentous appearance of the growth seemed consistent with a fungus. The use of small incubator tanks with untreated seawater during the incubation period likely resulted in the proliferation of the fungal growth on the embryos. The clutch left for the parents to guard in 2024 did not appear to succumb to any infection while under the care of the male, potentially due to a combination of parental care removing unviable eggs and higher water volume and exchange rate to dilute harmful microorganism growth. Future attempts to incubate eggs should test the effectiveness of disinfecting the eggs using, for example, formalin (Francis-Floyd, 1996; Rach et al., 1997) or ozone (Powell & Scolding, 2018). Tumble culture, similar to methods developed for sturgeon egg incubation (Gisbert & Williot, 2002), could also prove an effective method to keep the eggs properly aerated and clean during development.

Eggs in the centre of the egg masses were observed to develop more slowly, which led to the 8-day hatching period. While not conclusively investigated, it is likely the eggs developed more slowly in the centre of the mass because of lower oxygen and water circulation, which has been observed in lingcod egg clutches (Giorgi & Congleton, 1984). In addition to using egg tumblers, dividing the egg mass into smaller portions may result in more successful and synchronous hatching.

During incubation, fish eggs possess two primary energy sources: yolk and oil globules. Yolk serves as the primary energy source to support growth and development, while oil globules provide additional

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energy and buoyancy (Kamler, 2002). Both energy sources were tracked throughout development of embryos in this study by measuring the volumes of the yolk sac and oil globule (Figure 3). The yolk was primarily used for early cell division and development, being depleted by approximately 90% after 10 DPF. The oil globule became the primary energy reserve for the remainder of embryonic development until hatching when both yolk and oil globule energy reserves were less than 5%.

Within hours of hatching, larvae were mobile in the water column, able to swim in short bursts but struggled to maintain constant swimming for more than a few seconds. During this larval period, fish species continued to develop and grow, with the transition from endogenous to exogenous feeding being another key phase in the early life history of many aquatic species and aquaculture husbandry (Hjort, 1914; Kiørboe et al., 1985). Endogenous energy reserves were quickly depleted (within 1 day) and by the second day, larvae were actively searching for food. It became apparent live food needed to be immediately and consistently present post hatch or substantial mortalities occurred within 1 or 2 days of yolk and oil reserve depletion. High mortality during the onset of exogenous feeding is well documented in many species both in nature and laboratory conditions, and *C. violaceus* were not an exception (Anderson, 1988; Garrido et al., 2015; Yúfera & Darias, 2007).

Reproductive biology and parental care have been examined in other related species. Smaller species, such as *Anoplarchus purpurascens*, *Ulvaria subbifurcata* and *Xiphister atropurpureus*, produced eggs of similar size as *C. violaceus*, but in much smaller clutches of several thousand, with the larger *Xiphister mucosus* producing larger clutches averaging 7039 eggs (Coleman, 1992; LeDrew & Green, 1975; Marliave & DeMartini, 1977). Interestingly, many similarities were observed between this study and a report by Marliave (1987) on captive spawning of the wolf eel *Anarrhichthys ocellatus*. While specific metrics such as clutch size, egg size and spawning season differ, some behaviours seem similar enough that successful spawning and egg-rearing techniques used for the *A. ocellatus* could be implemented for *C. violaceus*. Marliave (1987) noted aggression was commonly observed between males to defend nesting dens from potential rivals. It was also noted that eggs removed from the parents had difficulties hatching, similar to what was experienced in this study. Massaging the egg masses by hand to mimic parental manipulations proved effective for enabling larvae to emerge from the eggs without issue, a technique which may be useful to implement with *C. violaceus* as well.

Despite limited success with captive breeding of *Cebidichthys violaceus*, results from this study will aid in further developing this species for aquaculture. It is recommended that future attempts at breeding this species should use one male per tank, remove egg masses immediately, separate masses into smaller clutches and experiment with various methods of aeration and water flow to keep the eggs free of fungus and debris. Observations of spawning behaviour and parental care from this study fill existing knowledge gaps for the largest herbivorous fish in California and provide another example of uniparental care in the family Stichaeidae. Courtship behaviours remain unknown and future tank designs should be designed to permit unobstructed observations inside spawning dens.

AUTHOR CONTRIBUTIONS

Luke Gardner and Scott Hamilton conceived the idea and secured funding for this study. Matthew Hoehn designed the study with assistance from Luke Gardner, Scott Hamilton and Kevin Hinterman. Matthew Hoehn and Kevin Hinterman conducted experiments, collected data and performed statistical analyses. Kevin Hinterman drafted the manuscript. Luke Gardner, Scott Hamilton and Matthew Hoehn revised and approved the manuscript.

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